VLSI Devices
Lecture 6

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L6

MOS equation

- Up to now, $Q_s(\phi_s)$ is found. We can control only V_g.
  - Relation between V_g and ϕ_s

Total silicon charge per unit area

$$V_g - V_{fb} = V_{ox} + \phi_s = -\frac{Q_s}{C_{ox}} + \phi_s$$

Taur, Eq. (4.7)

Taur, Eq. (4.16)

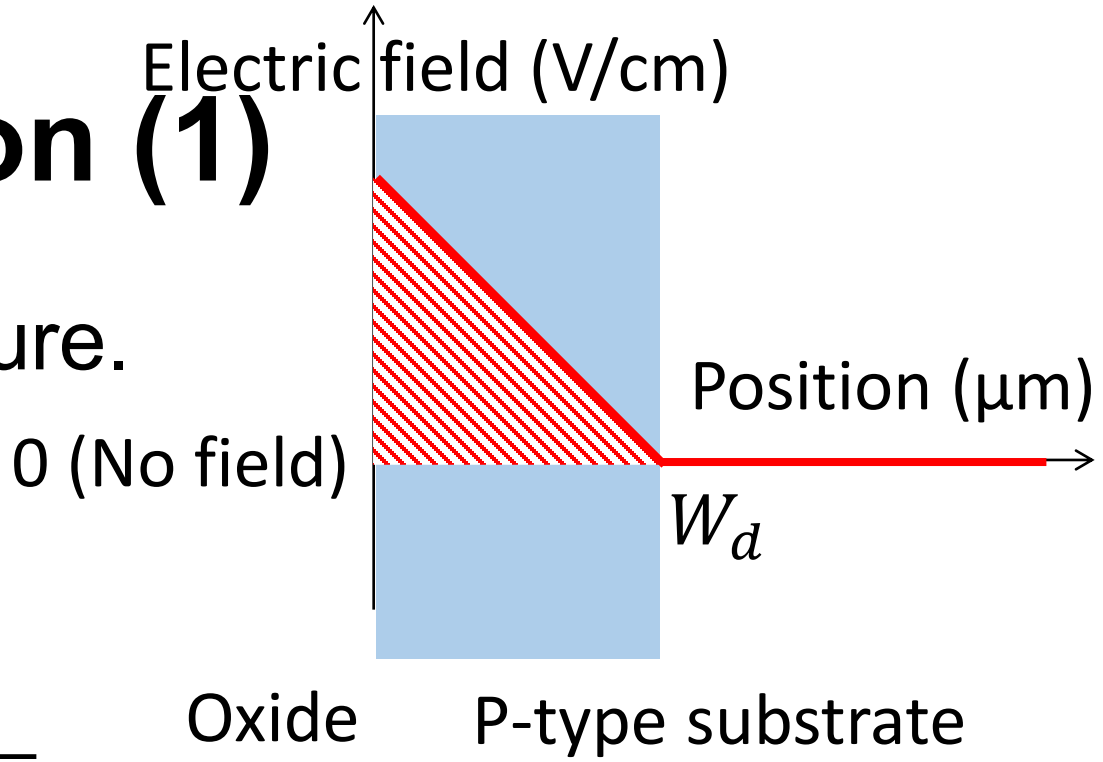
$\frac{\epsilon_{ox}}{t_{ox}}$, oxide capacitance per unit area

  - In general, $Q_s(\phi_s)$ is known. Now, we have

Taur, Eq. (4.31)

$$V_g - V_{fb} = \phi_s + \frac{\sqrt{2\epsilon_{si}k_BTN_a}}{C_{ox}}\left[\left(\exp\left(-\frac{q\phi_s}{k_BT}\right) + \frac{q\phi_s}{k_BT} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi_s}{k_BT}\right) - \frac{q\phi_s}{k_BT} - 1\right)\right]^{1/2}$$

Depletion approximation (1)



- Consider a depleted MOS structure.
  - With the depletion width, W_d,

$$\phi_s = \frac{1}{2} W_d \left(q \frac{N_a}{\epsilon_{si}} W_d \right)$$

  - Then,

$$W_d = \sqrt{\frac{2\epsilon_{si}\phi_s}{qN_a}}$$

Taur, Eq. (4.22)

  - Total depletion charge in silicon, Q_d, is

$$Q_d = -qN_aW_d = -\sqrt{2\epsilon_{si}qN_a\phi_s}$$

Taur, Eq. (4.23)

Depletion approximation (2)

- Another derivation from Eq. (4.16)
  - When $V_g > V_{fb}$, $\frac{d\phi}{dx} = -\sqrt{\frac{2k_BTN_a}{\epsilon_{si}}}\left[\left(\exp\left(-\frac{q\phi_s}{k_BT}\right) + \frac{q\phi_s}{k_BT} - 1\right) + \frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi_s}{k_BT}\right) - \frac{q\phi_s}{k_BT} - 1\right)\right]^{1/2}$
  - In the depletion region($2\phi_B > \phi_s > \frac{k_BT}{q}$), $\frac{d\phi}{dx}$ is approximated as

$$-\sqrt{\frac{2k_BTN_a}{\epsilon_{si}}}\left[\left(\cancel{\exp\left(-\frac{q\phi_s}{k_BT}\right)} + \frac{q\phi_s}{k_BT} \cancel{- 1}\right) + \cancel{\frac{n_i^2}{N_a^2}\left(\exp\left(\frac{q\phi_s}{k_BT}\right) - \frac{q\phi_s}{k_BT} - 1\right)}\right]^{1/2}$$

Depletion approximation (3)

- A good approximation in the depletion region

$$\frac{d\phi}{dx} = -\sqrt{\frac{2qN_a\phi}{\epsilon_{si}}}$$

Taur, Eq. (4.18)

  - Rearranged as

$$\frac{1}{\sqrt{\phi}}d\phi = -\sqrt{\frac{2qN_a}{\epsilon_{si}}}dx$$

Taur, ~Eq. (4.19)

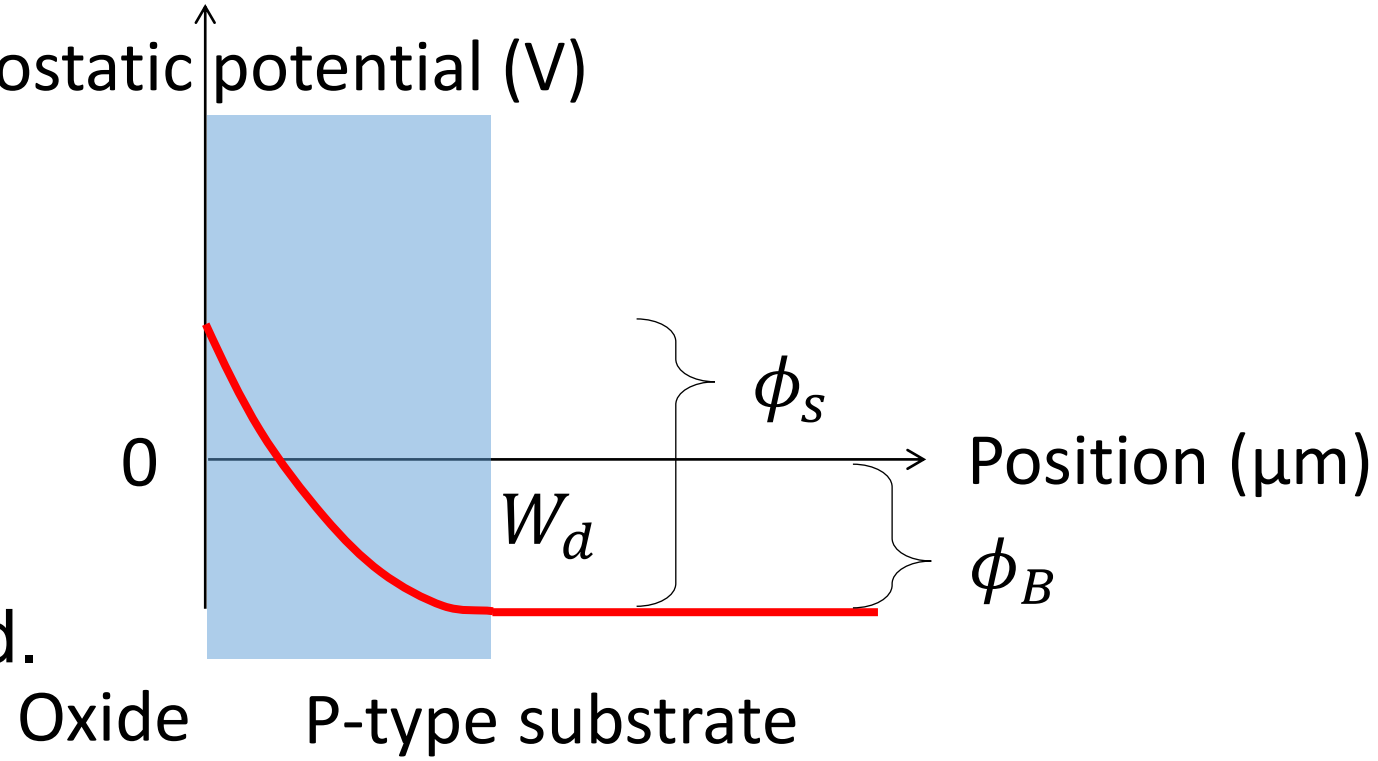
  - Integration from 0 to W_d yields

$$2\left(\sqrt{0} - \sqrt{\phi_S}\right) = -\sqrt{\frac{2qN_a}{\epsilon_{si}}}W_d$$

Taur, ~Eq. (4.22)

Potential profile

- A parabolic potential profile
  - The depletion region cannot grow indefinitely.
  - When $\phi_s = 2\phi_B$, $n(0) = p(\infty)$
  - The surface behaves like n-type material. (Inversion)
  - ϕ_s is approximately pinned.

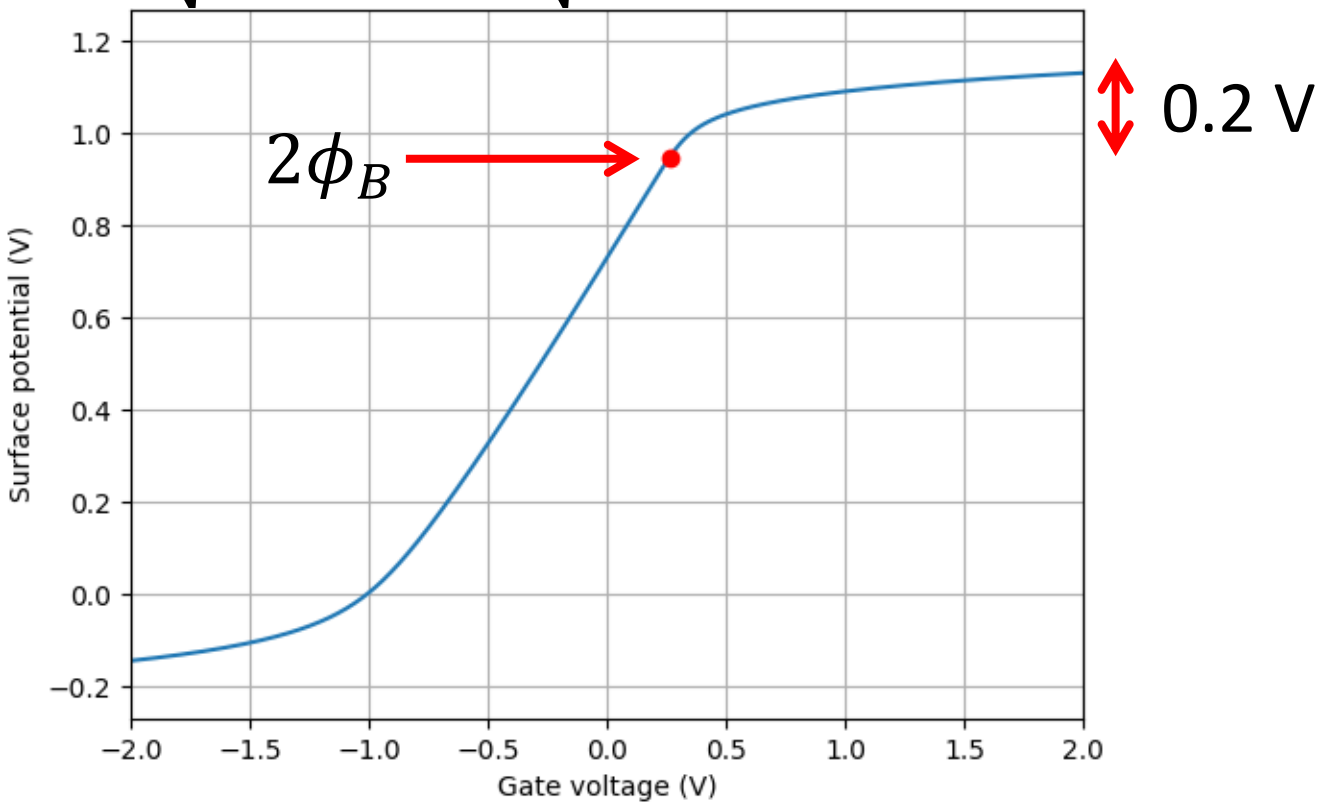


Maximum depletion width

- Therefore, maximum depletion width becomes

$$W_{dm} = \sqrt{\frac{4\epsilon_{si}\phi_B}{qN_a}} = \sqrt{\frac{4\epsilon_{si}k_B T\ \ln(N_a/n_i)}{q^2 N_a}}$$

Taur, Eq. (4.24)



Beyond threshold voltage

It's not perfectly fixed.

- The surface potential is *almost* fixed. (Surface potential pinning)
  - Small additional change in ϕ_s induces an exponential increase of the electron density.
  - Remember that $n = n_i \exp\left(\frac{q\phi}{k_B T}\right)$.
  - When $\phi_s = 2\phi_B$, (in other words, $\phi(0) = \phi_B$)

$$n(0) = n_i \exp\left(\frac{q\phi_B}{k_B T}\right) = p(\infty)$$

  - Additional potential ($\Delta\phi$) yields

$$n(0) = p(\infty) \exp\left(\frac{q\Delta\phi}{k_B T}\right)$$

It's a high density.

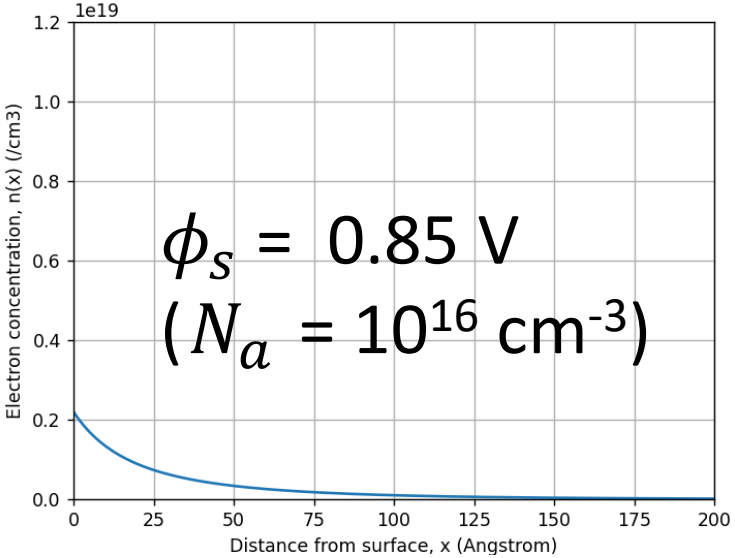
Strong inversion

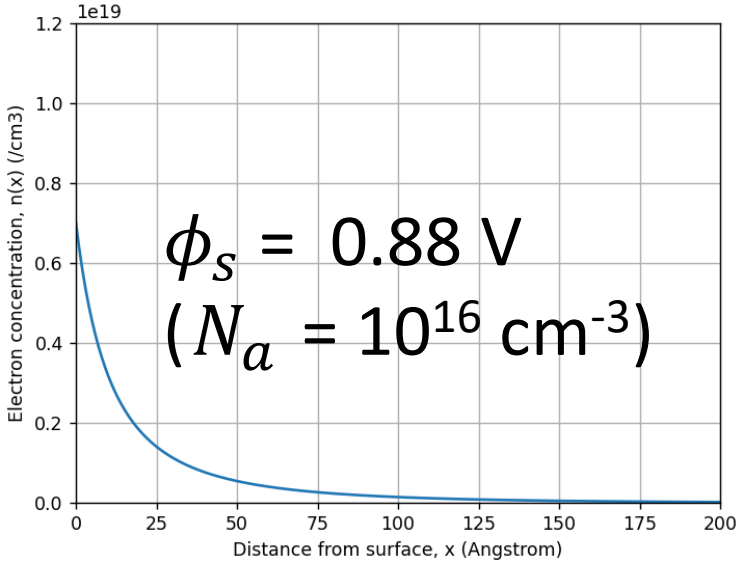
- Beyond strong inversion,

$$\frac{d\phi}{dx} \approx -\sqrt{\frac{2k_B T N_a}{\epsilon_{si}}\left(\frac{q\phi}{k_B T} + \frac{n_i^2}{N_a^2}\exp\left(\frac{q\phi}{k_B T}\right)\right)}$$

Taur, Eq. (4.25)

  - The electrons are distributed extremely close to the surface with an inversion-layer width less than 50 Å.



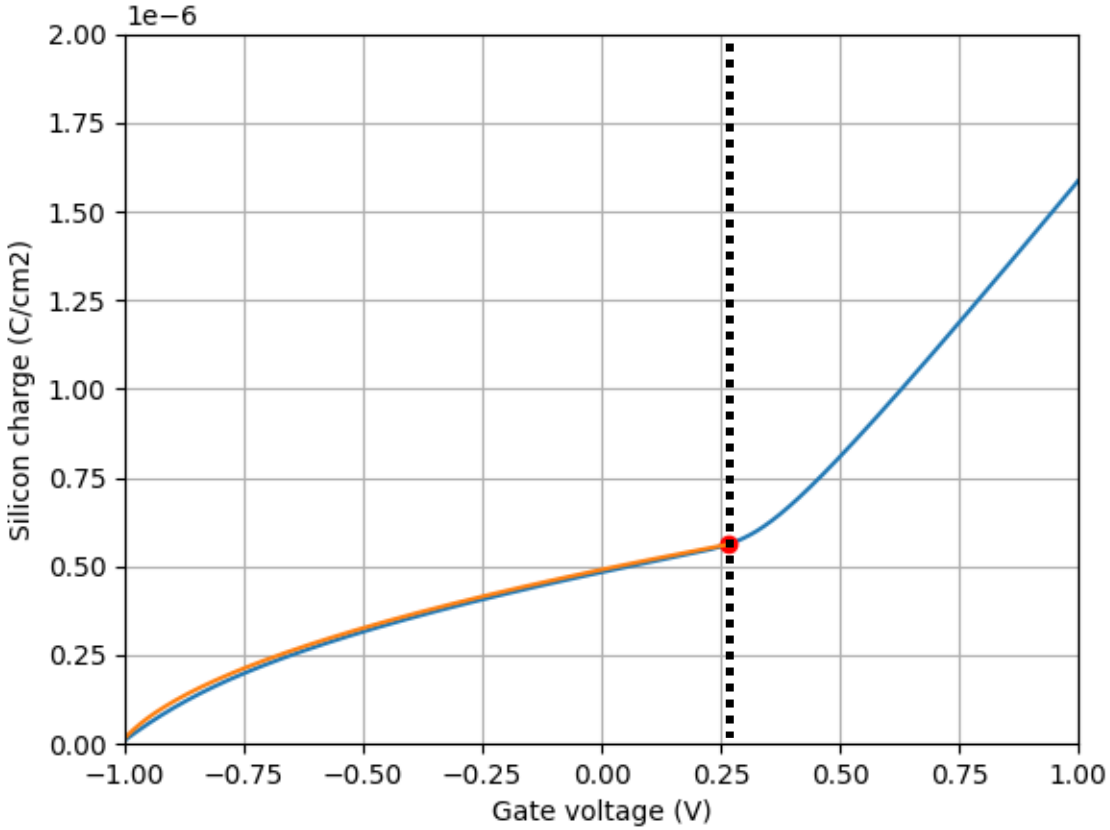


MOS equation with depletion approx.

- An approximate equation

$$V_g - V_{fb} = \frac{\sqrt{2\epsilon_{si} q N_a \phi_s}}{C_{ox}} + \phi_s$$

Taur, Eq. (4.32)



Thank you!